

Regression Algorithms for Learning & Its Application on Crypto Currency Markets

Report 2

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1. Abstract

This report presents the findings of a literature review done on the use of regression based and other machine learning models used for financial forecasting giving an idea of how such implementations can be used for making cryptocurrency predictions. Two main studies used are: Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research* by Fischer, T., & Krauss, C in 2018 and Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications* by Patel, J., Shah, S., Thakkar, P., & Kotecha, K. in 2015. These studies show how regression principles form the foundation for modern predictive systems that capture market dynamics, volatility, and non-linearity, characteristics which can all also be seen in the crypto market.

2. Introduction

Forecasting financial markets is a complex task due to the volatile nature, noise and behavioral influences. Traditionally, Regression models have provided a statistical framework for identifying relationships between input variables and future prices. With advances in computing and data availability, these models have evolved into more sophisticated machine learning systems which can learn complex dependencies.

This report reviews two such works that apply regression-inspired models to financial data. In 2018, Fischer and Krauss published their findings of using deep learning methods to capture sequential patterns in the S&P 500 market, while in 2015, Patel, Shah, Thakkar, and Kotecha published their exploration of feature-based regression models on Indian stock indices. Their findings contribute directly to the understanding of cryptocurrency price prediction, where similar data irregularities and non-stationarity exist.

3. Fischer & Krauss (2018): LSTM for Financial Time Series

In 2018, Fischer and Krauss published their investigation of the application of Long Short-Term Memory (LSTM) networks to predict daily price movements of S&P 500 stocks between 1992 and 2015. Long Short-Term Memory (LSTM) networks are a recurrent neural network (RNN) architecture. Their goal was to predict whether a stock would outperform or underperform the market median on the next day.

The authors constructed a dataset covering all S&P 500 constituents from Thomson Reuters making sure there was no survivor bias. Each study period contained 750 trading days for training and 250 days for testing. For every stock, the model used sequences of 240 consecutive daily returns which were then normalised using mean and standard deviation from the training period. These formed the input features while the binary target variable indicated upward or downward movement. Although an LSTM is a deep learning architecture, it still relies on regression fundamentals. Each LSTM cell applies weighted linear transformations followed by non-linear activations which enables the model to learn complex relationships between lagged inputs. The use of regularisation methods such as dropout and early stopping is quite similar to the concept of ridge regression as it helps prevent overfitting on volatile data.

To assess performance, Fischer and Krauss compared the LSTM with three benchmarks: Random Forest (RAF), Deep Neural Network (DNN), and Logistic Regression (LOG). Logistic regression provided a linear baseline while RAF and DNN represented non-linear alternatives. The LSTM model results showed the highest predictive accuracy by producing average daily returns of 0.46% and a Sharpe ratio of 5.8 before transaction costs. It consistently outperformed the other models as RAF produced 0.43% daily returns while DNN and LOG produced 0.32% and 0.26% returns respectively. The LSTM's advantage came from its ability to model temporal dependencies and reversal patterns thus identifying stocks with short-term overreactions that were later corrected.

Fischer and Krauss's study shows how regression based learning has been effectively applied to financial markets filled with volatility and temporal dependence. Their study extends regression principles into deep learning architectures which enables the modelling of complex non-linear relationships in stock data. Although limited to traditional markets, this research is an example of the evolution of financial forecasting from linear models to sequential frameworks. It offers valuable insight into how similar approaches based on regression may be useful in the study of volatile markets such as cryptocurrencies.

4. Patel, Shah, Thakkar, Kotecha(2015): Trend Based Regression & Data Preparation

In 2015, Patel, Shah, Thakkar, and Kotecha published their exploration in which they analysed ten years of Indian stock market data between 2003–2012 from two major indices (CNX Nifty and S&P BSE Sensex) and two stocks (Reliance Industries and Infosys Ltd.). Their objective was to examine the effectiveness of various machine learning techniques inspired by regression for predicting the direction of price movement.

The authors introduced two approaches for preparing input data. The first used continuous valued technical indicators such as the Simple Moving Average, Relative Strength Index (RSI), Momentum, and Commodity Channel Index (CCI). These indicators act as lagged variables in a regression framework by quantifying the relationship between past and future prices. The second approach called Trend Deterministic Data Preparation (TDDP) transformed these continuous indicators into +1 (uptrend) or -1 (downtrend) signals. This method effectively converted continuous regression style input into a categorical format which would be suitable for classification. Hence, the study examined how transforming numerical data into discrete directional trends could influence model accuracy and interpretability.

The four algorithms implemented were: Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forest (RF), and Naïve Bayes. Each of them was trained on 50% of the dataset and tested on the remaining 50%. The following were the findings:

- Using continuous-valued data, the Random Forest model performed best overall.
- When using the trend deterministic approach, all models achieved notably higher accuracy.

Data preprocessing and feature representation are key principles originating from regression analysis. These findings underscored their role in determining predictive performance. The Random Forest, in particular, benefited from ensemble averaging, which can be viewed as a collection of regression trees fitted to various subsets of data to enhance generalisation.

This work serves as an important example of how regression and machine learning methods have been applied in financial forecasting. Their study demonstrates that the preprocessing of input features can significantly affect the success of predictive models. This research establishes a methodological approach for the preprocessing, modelling, and evaluation of stock price data. It contributes to the broader literature demonstrating the adaptability of regression-based frameworks to different financial environments.

5. Comparing the Two Works

Both works look into the predictive capabilities of regression related models within financial markets but differ in their methodological approach. Fischer and Krauss extend regression into a temporal sequence-learning framework using LSTM networks, while Patel, Shah, Thakkar, and Kotecha examine feature based transformations and ensemble methods to improve prediction accuracy.

Aspect	Fischer & Krauss LSTM Regression for Financial Time Series	Patel, Shah, Thakkar, & Kotecha Trend Based Regression & Data Preparation
Dataset	S&P 500 daily returns (1992–2015)	Indian stocks and indices (2003–2012)
Model Type	LSTM (sequential regression)	ANN, SVM, RF, Naïve Bayes
Key Contribution	Demonstrates effectiveness of memory based models for market sequences	Shows impact of trend based data encoding
Connection to Regression	Regularised temporal modelling	Feature engineering and ensemble regression
Core Outcome	Outperformed benchmarks in accuracy and returns	Improved results through TDDP preprocessing

These studies document how regression has evolved in financial modelling from linear and static estimation to dynamic, data driven systems that capture complex dependencies. Both demonstrate that although the specific techniques may differ, the fundamental principle of learning relationships between predictors and target variables remains at the core of it. These studies therefore provide an essential theoretical and empirical background for understanding how models based on regression have been applied across financial domains.

6. Conclusion

The reviewed works highlight the progression of regression based approaches in financial forecasting. Fischer and Krauss demonstrate how integrating regression concepts into recurrent neural networks allows for the modelling of time dependent structures in market data. Patel, Shah, Thakkar, and Kotecha, on the other hand, show that preprocessing and transformation of input variables, which are key aspects of regression design, significantly affect predictive outcomes.

Both studies contribute to the historical and methodological background of financial prediction research. They collectively demonstrate that regression, in both its linear and non-linear forms, remains a foundational tool in capturing financial relationships. These bodies of work provide valuable context for contemporary forecasting studies, including those examining modern data sources such as cryptocurrencies, where similar challenges of volatility and uncertainty persist.

7. Bibliography

1. Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.
<https://www.sciencedirect.com/science/article/abs/pii/S0377221717310652>
2. Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015). Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications*, 42(1), 259–268.
<https://www.sciencedirect.com/science/article/abs/pii/S0957417414004473>